

MICROSOFT SQL SERVER

Enhance your SQL skills and gain the competitive edge



Module 1: Introduction to Relational Databases and SQL

- Introduction to Databases and Database Management Systems (DBMS)
- Relational Database Concepts
 - Normalization Concepts (1NF, 2NF, 3NF)
 - Denormalization and its Use Cases
- Introduction to SQL (Structured Query Language)

Module 2: Database Design

- Database Design Principles
- Entity-Relationship (ER) Modelling
- Indexes and Keys in SQL Server
- SQL Data Types and Operators

Module 3: SQL Server Installation and Configuration

- Installing SQL Server
- SQL Server Configuration Options
- SQL Server Management Studio (SSMS) Overview
- Connecting to SQL Server Instances

Module 4: Data Definition, Manipulation and Transaction Control

- DDL Commands
 - Create, Alter, Drop, Truncate
- DML Commands
 - Select, Insert, Update, Delete
- DCL Commands
 - Grant, Revoke
- TCL Commands
 - Commit, Rollback, Save point

Module 5: Working with SQL: Join, Functions, and temp table

- Combining rows from tables using inner, outer, cross, and self joins
- Deploying operators such as intersect, except, union
- Understanding SQL functions and their usage
- List of built-in SQL Functions
 - Aggregate functions
 - Date & Time functions
 - String functions
- Temporary table creation

Module 6: SQL Views and Stored Procedures

- Learning SQL views
 - Advantage of views
 - Managing views in SQL Server
- Understanding stored procedures and their key benefits
- Working with Stored Procedures
 - Executing a SP with Parameters
 - Executing a SP without Parameters

Module 7: SQL Indexes and Triggers

- Creating indexes & their guidelines
- Clustered index Non-Clustered index
- Creating DML Triggers
- Instead of Triggers
- Enable & disable DML Triggers

Module 8: Real-world Projects and Case Studies

- Hands-on SQL Server Projects Case
- Studies from Industry
 - Best Practices and Industry Standards

Module 9: Revision and Certification Preparation

- Review of Key Concepts
- Practice Tests and Quizzes
- Tips and Strategies for Certification Exams (e.g., Microsoft Certified: Azure Database Administrator Associate)

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